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RESEARCH ARTICLE



A spatiotemporal knowledge graph for driving factor analysis and growth prediction of regional economy in China

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ABSTRACT

Knowledge Graphs (KGs), as a tool for organizing and integrating multidimensional data, have become increasingly important in addressing multifaceted challenges in geographic research. However, most existing KGs rely on static data to organize information, overlooking the dynamic and spatiotemporal properties inherent in real-world scenarios, thereby resulting in incomplete representations of spatiotemporal knowledge. Similarly, traditional economic analysis methods struggle to unravel the complex relationships among numerous factors. Thus, to address these limitations, we integrated multidimensional spatiotemporal data and combined the complex relationships between multiple factors of economic activities to construct a regional economic spatiotemporal knowledge graph (REST-KG), enabling the modeling of systemic economic interactions. Utilizing relational graph convolutional networks (R-GCNs), we trained knowledge representations on REST-KG to further identify the main driving factors of economic growth and predict future trends in key economic indicators. Our findings reveal that the primary drivers of China's economic growth exhibit temporal variations across different periods. Notably, night lights and employment emerge as the most significant factors influencing GDP growth from 2001 to 2021. Concurrently, the influence of traditional statistical indicators on GDP is progressively diminishing. Our prediction results for China's average annual GDP growth rate from 2022 to 2025 (with absolute error under 0.3%) show that China's economy will maintain stable and sustained growth. Furthermore, economic inequality, marked by inter-city disparities, narrowed significantly from 2000–2012 but has since widened slowly, with projections of further increases without targeted policies. An external shock module enhances predictions under disruptions. These findings indicate that China's economy is progressing toward balanced and sustainable development, yet targeted policy interventions remain essential to achieve further common prosperity.

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1 Introduction

Spatiotemporal Knowledge Graphs (STKGs) are a powerful data representation technology that integrates multi-source, heterogeneous data to model dynamic relationships across space and time (Chen et al. 2022; Sun, Xu, and Hu 2022), surpassing static traditional knowledge graphs. By integrating spatial heterogeneity and temporal dynamics, STKGs have become instrumental in domains such as urban planning, transportation, and environmental analysis (Ge et al. 2022). In regional economic analysis, STKGs offer significant potential to address the complexity of economic systems characterized by diverse, evolving activities and inter-city interactions driven by policy, trade, and innovation (Ning et al. 2023). However, constructing STKGs to model these intricate spatiotemporal economic interactions, accurately predict future trends, and uncover key economic drivers remains a formidable challenge, necessitating advanced methodologies to navigate the multifaceted dynamics of regional economies.

Regional economies exhibit both systemic wholeness and internal complexity (Berry 1964), where individual cities function as systems and the network of cities forms a larger interconnected system

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(Chugunov et al. 2021; Ruth and Hannon 2012). Intra-city economic systems encompass interdependent activities such as employment, investment, and consumption, shaped by local resources, innovation, and policy interventions. Concurrently, inter-city economic systems arise from spatial and economic interactions, including trade, migration, and infrastructure connectivity, which collectively drive regional dynamics in response to globalization and policy shifts (Alves et al. 2015; Turkina, Van Assche, and Kali 2016). Unlike existing spatiotemporal knowledge graphs that focus on single entities and activities, such as traffic flow, the regional economic spatiotemporal knowledge graph integrates diverse entities (e.g. cities, economic indicators) and multidimensional relationships to model the dynamic economic interactions within complex urban systems, thereby addressing the challenges of regional economic modeling.

Knowledge graph embedding, a core component of STKGs, embeds entities and relationships into low-dimensional spaces, preserving semantic and structural information (Scarselli et al. 2008; Schoenmackers et al. 2010). Early models like TransE used translational approaches but struggled with complex, multi-relational structures (Bordes et al. 2013). Advancements such as DistMult's bilinear operations improved relational modeling (Yang et al. 2014). Graph Neural Networks (GNNs), including Graph Convolutional Networks (Kipf 2016), Graph Attention Networks (Veličković et al. 2017), and Relational Graph Convolutional Networks (Schlichtkrull et al. 2018), further capture spatiotemporal interactions. Recent advancements, such as multi-scale dynamic convolutional networks (Zhang et al. 2020), multi-granularity relational augmentation networks (Xue et al. 2023), and multi-perspective hierarchical relation networks (Xue et al. 2025), have enhanced embedding capabilities for dynamic systems. These embeddings support downstream knowledge reasoning such as EDMF, CARM, and multi-view recommendation (Li et al. 2021; Liu et al. 2021; Lu et al. 2016). For regional economic research, we can predict future economic trends and analyze the driving forces of economic development to gain a deeper understanding of economic dynamics (Lin et al. 2015; Rossi et al. 2021).

Knowledge reasoning within Spatiotemporal Knowledge Graphs (STKGs) significantly enhances regional economic analysis by overcoming the limitations of traditional econometric models, such as Vector Autoregression (VAR) and spatial panel regression (Easterly and Rebelo 1993; Førsund 1985; Phillips 1996), which struggle with non-statistical indicators and nonlinear relationships (Fan and Sun 2008). In contrast, reasoning techniques based on graph-based Graph Neural Networks excel at uncovering hidden economic patterns (Scarselli et al. 2008), such as the impact of policy interventions on GDP or the ripple effects of infrastructure investments across regions. These methods enable precise forecasting and support informed policy-making, leveraging the systemic interconnectedness of regional economies to promote balanced development, offering a significant advancement over conventional approaches.

In summary, the regional economic spatiotemporal knowledge graph integrates diverse economic indicators into a cohesive system theory framework, modeling regional economies as interconnected networks of complex urban systems. By capturing nonlinear spatiotemporal interactions, STKGs enhance the accuracy of economic trend predictions and provide actionable insights for sustainable and equitable economic policies, aligning with China's goals for balanced regional development.

2 Method

2.1 Framework overview

This study was mainly divided into three steps: (1) Constructing the regional economic spatiotemporal knowledge graph (REST-KG). (2) Conducting knowledge representation learning on the REST-KG, using relational graph convolutional networks (R-GCNs) for model training, and obtaining entities embeddings. (3) Performing knowledge reasoning based on the entities embeddings results: potential relationship identification and future economic indicator value prediction. The entire task can be framed as a progression from “data” to “information” to “knowledge.”

2.2 REST-KG construction

2.2.1 Definition of knowledge graph

A Knowledge Graph (KG) is defined as a graph $G = (\mathcal{E}, \mathcal{R}, \mathcal{F})$, consisting of the entity set $\mathcal{E}$, relation set $\mathcal{R}$ and fact set $\mathcal{F}$. Each fact in $\mathcal{F}$ is a triplet (h, r, t) , where $h, t \in \mathcal{E}$, $r \in \mathcal{R}$, denoting a directional edge from

head entity h to tail entity t with relation type r . Each entity can have an entity type determined by a mapping function $\phi: \mathcal{E} \rightarrow \mathcal{A}$, where $\mathcal{A}$ is the set of predefined entity types. The KG schema describes the high-level structure of KG, which shows the types of entities and relations between entity types (Hogan et al. 2021; Ji et al. 2021) (Table 1).

2.2.2 Schema of REST-KG

The entities in the REST-KG were categorized into three types: spatial entities (e.g. cities), temporal entities, and indicator entities (e.g. GDP entities and population size entities). Raw data are mapped into the attributes of 25 different indicator entities (Figure 1c). Spatial entities are connected to indicator entities through spatial associations. At the same time, indicator entities are linked to temporal entities according to the year they belong to. Spatial entities are connected by “Nearby” relationships, while temporal entities are linked by “Earlier than” relationships. These entities, relationships, and their attributes together form the basic framework of REST-KG (Figure 1b).

2.2.3 Data of REST-KG

This study took Chinese counties as an example to construct the regional economic spatiotemporal knowledge graph (REST-KG). Traditional statistical data comes from the *Statistical Yearbook of China* and *China County Statistical Yearbook*, including 74 indicators such as GDP, construction land area, and total urban population. We handle missing values in the statistical yearbook by linear interpolation to maintain temporal continuity. The night light data was sourced from the 2000–2021 global 500-meter resolution NPP-VIIRS-like night light data set (Chen et al. 2020). We also collected data on extreme climate to enrich our knowledge graph.

Non-statistical indicators, such as nighttime light and extreme climate data, require specialized processing for integration. Nighttime light data from NPP-VIIRS are preprocessed to address outliers by setting a threshold based on maximum light intensity in fast-developing regions (e.g. Beijing, Shanghai), capping pixels exceeding this value. Zonal statistics compute average and total nighttime light intensity of each county. Extreme climate data are aligned with counties by mapping site coordinates to administrative boundaries, calculating annual counts of extreme rainfall and high-temperature days. All data are normalized using min-max scaling to map values to the range to mitigate scale differences.

2.3 REST-KG embedding

2.3.1 Relational graph convolutional networks

This study employed unsupervised Relational Graph Convolutional Networks (R-GCNs) with a negative sampling loss function (Schlichtkrull et al. 2018). R-GCNs extends traditional GCNs to handle graphs with multiple edge types, ideal for modeling knowledge graphs and relational data. Unlike standard GCNs, which assume homogeneous graphs, R-GCNs process heterogeneous edge types by incorporating relation-specific weight matrices, enabling learning of diverse relationship effects on node representations. Mathematically, an R-GCNs layer is formulated as:

$$H_v^{(l+1)} = \sigma \left(\sum_{r \in R} \sum_{u \in N_v^r} \frac{1}{c_{uv}^r} W^r H_u^{(l)} \right), \quad (1)$$

where $H_v^{(l+1)}$ represents the updated node feature for node v at layer $l + 1$, W^r is the relation-specific weight matrix for relation r , and N_v^r denotes the neighbors of node v connected by relation r . c_{uv}^r is the normalization factor and σ is the activation function.

Table 1. Relationships of REST-KG.

Category	Entity	Relationship
Spatial relationships	Spatial entity – spatial entity	Near by
Time relationship	Time entity – time entity	Earlier than
Indicator relationship	Indicator entity – spatial entity	Belongs to
	Indicator entity – time entity	At time
	Indicator entity – indicator entity	Promote, inhibit, constitute

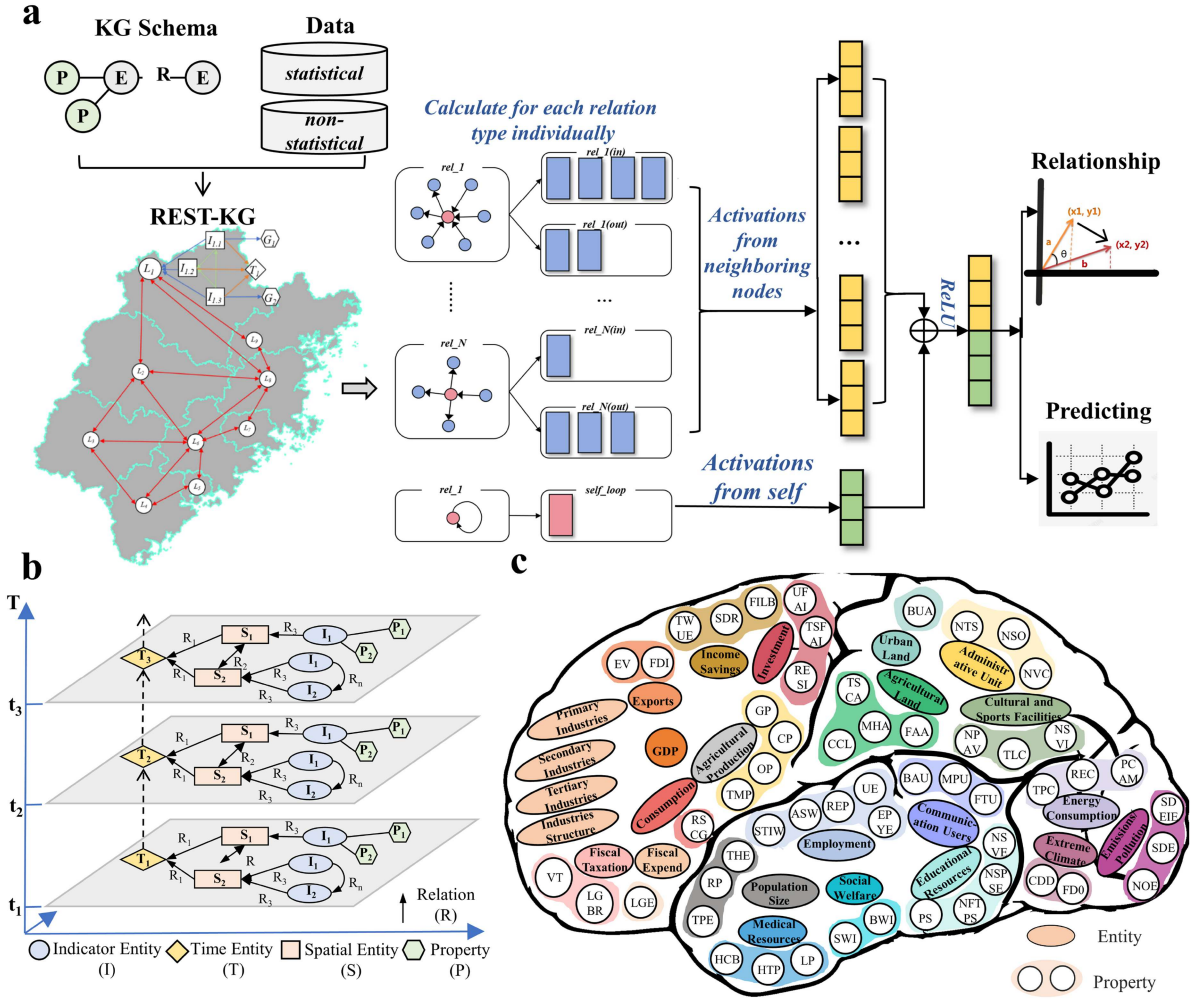


Figure 1. Framework. (a) Process of regional economic spatiotemporal knowledge graph (REST-KG) construction, embedding and knowledge reasoning. (b) Scheme of REST-KG. (c) Data of REST-KG. The abbreviations of the data are shown. For detailed descriptions of the specific data corresponding to these abbreviations, see Supplementary Table S1.

Table 2. The best configuration of R-GCNs embedding.

Parameter	Optimal configuration	Parameter	Optimal configuration
R-GCNs layers	2	L2 regularization	0.15
Hidden dimension	128	Neighbor sampling	(5,3)
Dropout rate	0.2	Batch size	512
Learning rate	0.001	Epoch	80

2.3.2 Hyperparameter selection

To optimize the performance of the relational graph convolutional networks (R-GCNs) model, we employed a systematic hyperparameter tuning process to identify the optimal configuration for our regional economic spatiotemporal knowledge graph (REST-KG). We conducted a combination of grid search and random search to explore a defined hyperparameter space, followed by Bayesian optimization to refine the selection. The best configuration after optimization selection is show in Table 2.

2.4 REST-KG reasoning

2.4.1 Potential relationship identification

We quantified the relationship strength between the target entities and other type of entities using cosine similarity. The cosine similarity between two vectors is computed as:

$$\text{Cosine similarity } (v_{\text{target}}, v_{\text{other}}) = \frac{v_{\text{target}} \cdot v_{\text{other}}}{\|v_{\text{target}}\| \cdot \|v_{\text{other}}\|}, \quad (2)$$

where v_{target} and v_{other} are the embedding vectors for target entities and other type of entities, respectively. The dot product measures the similarity between the vectors, and the denominator normalizes the vectors, ensuring the cosine similarity falls within the range of $(-1, 1)$.

Higher similarity indicates that the entity is semantically closer to the target entity, reflecting a stronger relationship (Park, Lee, and Yu 2024; Xue et al. 2021). We can rank all entities based on their cosine similarity to the target entity and identified the top entities with the highest similarity scores as those with the most significant influence. By sorting entities in this way, we can select the most relevant entities as key determinants of the target entity's behavior. This approach provides a clear and interpretable measure of entity relationships within the knowledge graph.

2.4.2 Indicator prediction

As an efficient and widely used machine learning algorithm, eXtreme Gradient Boosting (XGBoost) performs supervised learning tasks such as classification and regression by building an ensemble of decision trees (Alim et al. 2020; Scarselli et al. 2008). The XGBoost model formula is:

$$\hat{y}_i^{(t)} = \sum_{k=1}^t f_k(x_i) = \hat{y}_i^{(t-1)} + f_t(x_i), \quad (3)$$

where $\hat{y}_i^{(t)}$ is the result of sample i after the t^{th} iteration, $\hat{y}_i^{(t-1)}$ is the result of the first $t-1$ trees, $f_t(x_i)$ is the model function of the t^{th} tree.

Using KG embeddings, we extract node features to predict future city economic indicators, treating the task as a time series regression problem (Chen and Guestrin 2016). We apply a sliding window approach to construct and partition the dataset for XGBoost, with each window using 8 years of historical data (2001–2021) to predict the next 4 years, advancing by 1-year steps. The dataset is split into: a training set (2001–2016, predicting 2009–2020), a validation set (2010–2017, predicting 2018–2021), and a test set (2013–2021, forecasting 2022–2025). XGBoost parameters are detailed in Table 3.

3 Result

3.1 REST-KG construction and knowledge embedding

We defined cities, timepoints, and socioeconomic indicators as entities in REST-KG. Specifically, there are 6930 city entities, 21 timepoints entities, and 138,600 socioeconomic entities in REST-KG. Each entity has initial features, such as the GDP entities has initial features: GDP value, GDP growth, GDP growth rate (Figure 2a).

We then built relationships through proximity between cities, time sequence, the affiliation of indicators with cities, and the mutual influence between indicators. The REST-KG leveraged the proximity between cities to construct spatial relationships, enabling the aggregation and analysis of spatial information. As depicted in Figure 2b, the proximity between cities is visualized, revealing the strength and extent of spatial connections and interdependencies across regions. A Pearson correlation coefficient threshold of > 0.4 is used to identify relationships between indicators (Figure 2c–f), and these correlations are validated against existing literature to ensure robustness (Herrerias and Orts 2011; Wang and Hu 2007). The relationships between indicators are divided into facilitating, inhibiting, and constitutive relationships (Figure 2g).

Table 3. The parameters of XGBoost.

Parameter	Optimal configuration	Parameter	Optimal configuration
Max depth	6	L1 regularization	0.25
Learning rate	0.005	L2 regularization	1.2
n estimators	500	random state	42
Gamma	0		

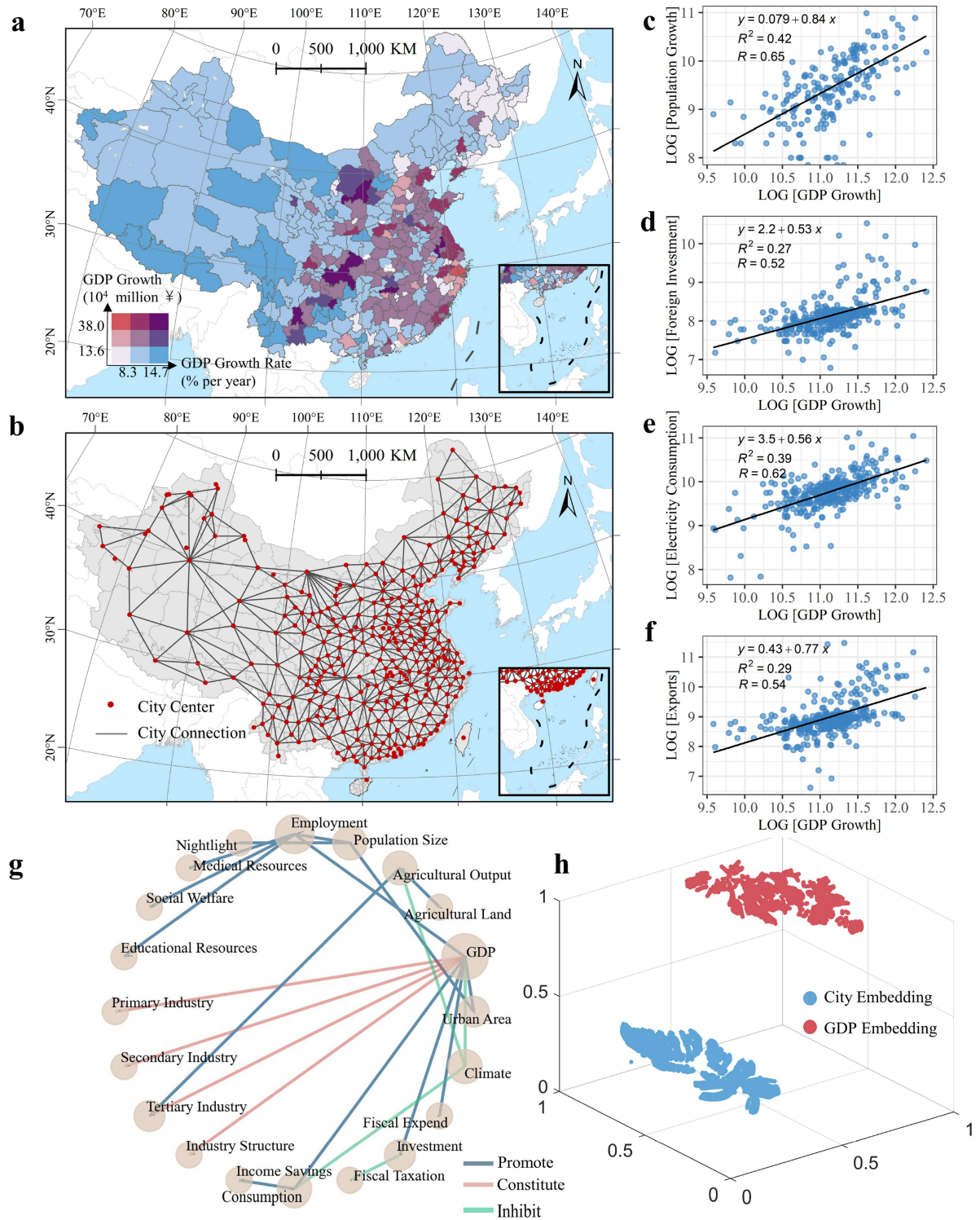


Figure 2. REST-KG entity, relationship, and knowledge embedding results. (a) GDP changes in 2001–2021. (b) City spatial relationship (determined by whether they are adjacent). (c)–(f) Indicators numerical relationship. (g) Indicator relationship construction. (h) Embedding results (T-SNE dimensionality reduction), taking GDP entities and City entities as examples.

After defining entities and building relationships, we established the REST-KG, and then we used R-GCNs to derive knowledge embeddings. As depicted in Figure 2h, the R-GCN model generated embedding results (vector representations) of entities that encapsulated the intricate interdependencies between entities and offered a nuanced understanding of the connections between economic factors, enabling more precise analysis and prediction of regional economic dynamics.

3.2 Driving factors of economic growth: a spatiotemporal analysis

Entity similarity, measured via cosine similarity, quantifies associations between socioeconomic indicators entities and GDP entities within the REST-KG. The comprehensive analysis spanning 2001 to 2021 reveals that nighttime light, employment, and consumption demonstrate exceptionally high similarity to GDP entities (over 0.8), establishing them as the primary targeted indicators influencing GDP growth (Figure 3a). Specifically, in 52%, 44%, and 41% of cities, nighttime light, employment, and consumption entities, respectively, exhibit similarities above the mean by more than one standard deviation (Figure 3b,c). To validate the robustness of these correlation-based findings, structural equation modeling (SEM) confirmed the significant positive impact of night lights, employment, consumption, and investment on GDP growth, thus strengthening the reliability of the identified drivers (Table 4).

The similarity between socioeconomic indicators and GDP exhibits a declining trend over time (Figure 3d). This trend is primarily attributed to shifts in the economic structure and the influence of emerging factors such as technological innovation, globalization, and policy adjustments. As new economic drivers continue to emerge, traditional indicators increasingly struggle to fully capture the dynamic evolution of GDP.

There are both homogeneity and spatial heterogeneity in the targeted indicators of GDP growth. In terms of homogeneity, more than 95% of the city target indicators are mainly concentrated in night lighting, employment and consumption (Figure 3e), reflecting the general importance of these indicators in

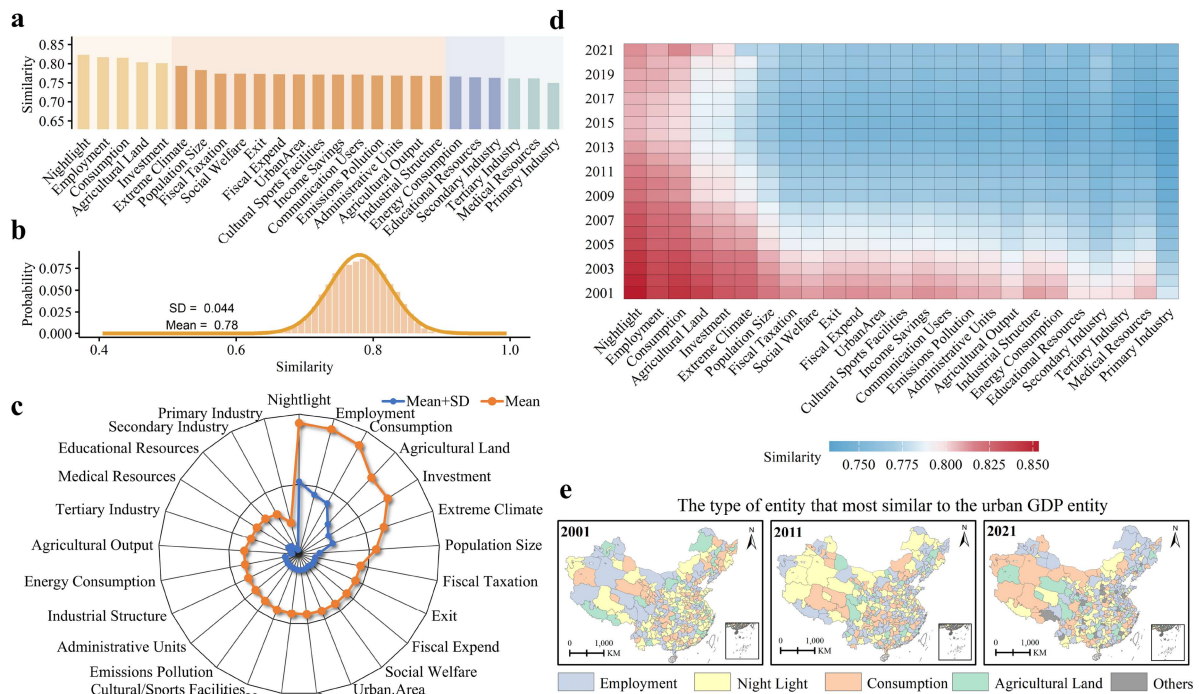


Figure 3. Entity similarity analysis results (calculated with GDP entity). (a) Average similarity of entities from 2001 to 2021. Identification of primary targeted indicators affecting economic growth. (b) Entities similarity probability distribution curve. The mean is 0.78 and the standard deviation (SD) is 0.044. (c) The proportion of entities of different types whose similarity is greater than the mean (and mean + SD). (d) Changes in similarity relationships between different entities and GDP entities over time. (e) The type of entities most similar to the GDP entities.

Table 4. SEM path coefficients.

Factor	Estimate	Std. error	p	Causal relationship with GDP growth
Nighttime light	0.0236	0.0032	<0.001	Positive effect
Employment	-0.0433	0.0064	<0.001	Effect
Investment	0.0890	0.0035	<0.001	Positive effect
Consumption	0.9334	0.0019	<0.001	Significant positive effect

promoting economic growth. However, target indicators exhibit a dispersed distribution across cities without forming distinct spatial clusters. This heterogeneity arises from diverse economic structures, development paths, and resource endowments, underscoring the necessity of locally tailored policies.

3.3 Economic indicator prediction

Using R-GCNs embeddings of GDP entities and key indicator attributes (nighttime light, employment, consumption, investment), we predict GDP growth via XGBoost. Validation against actual 2019–2022 GDP growth shows R-GCNs + XGBoost outperforming Random Forest, Linear Regression, ST-GCN, and LSTM, with lower MSE and MAPE (Figure 4c,d).

In addition, despite challenges capturing the 2020 GDP slowdown due to COVID-19, the R-GCNs + XGBoost model performed well in handling complex and nonlinear relationships. Although the pandemic was a sudden event, the long-term trends and structural relationships learned by the model during training are still valid. Therefore, in post-pandemic data, the model can continue to provide relatively accurate predictions.

The prediction results of GDP growth from 2022 to 2024 are presented in Figure 5a. Predicted GDP growth rates are 3.26% (2022), 5.49% (2023), and 4.99% (2024), 4.39% (2025) with errors within 0.3% of official reports (Table 5). It is worth noting that in these four years, the overall GDP growth rate showed a trend of being higher in the western and southeastern regions and lower in the northeastern region.

Analysis of compound annual growth rate (CAGA) across cities of varying scales reveals that all cities experienced rapid growth from 2000–2010, followed by a decline, with average annual growth (AAG)

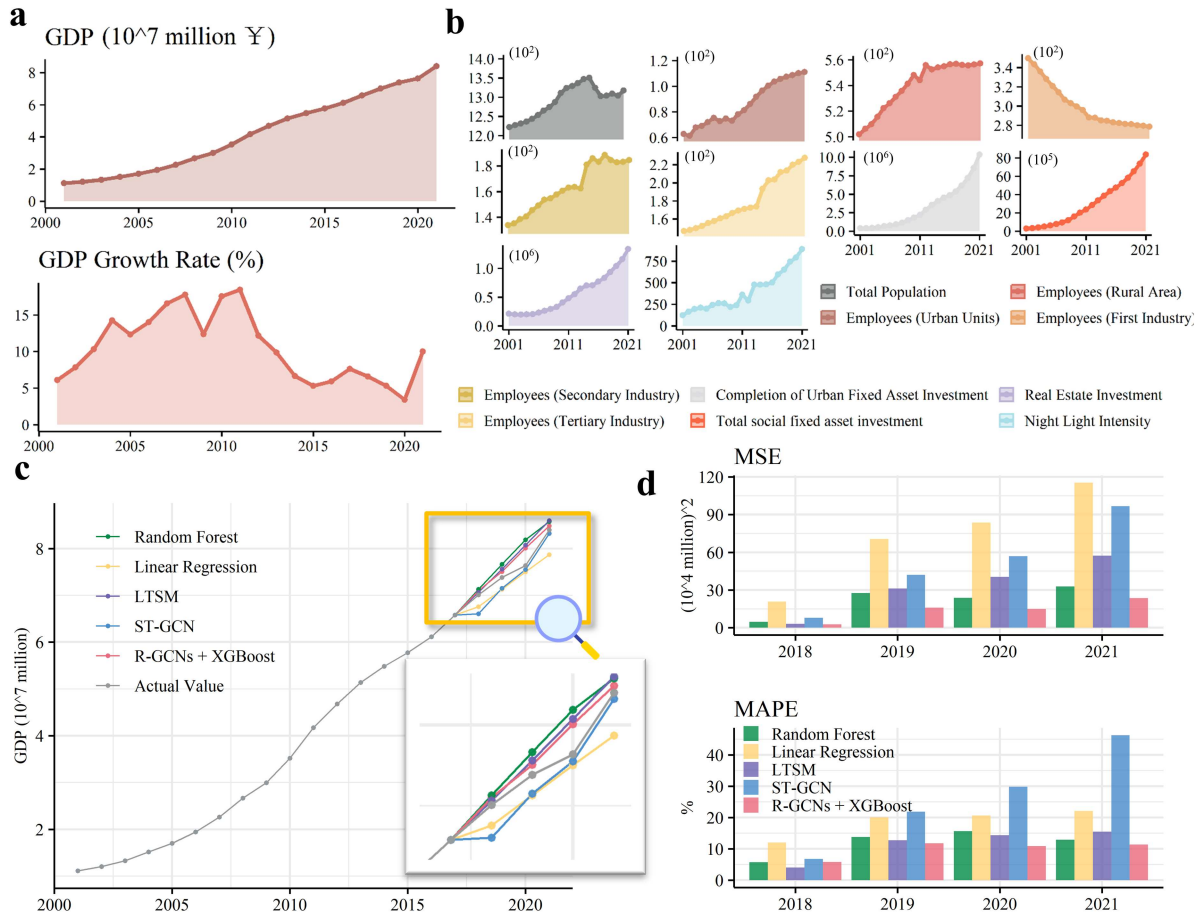


Figure 4. GDP prediction model and accuracy assessment. (a) GDP and GDP growth rate in 2010–2021. (b) Primary targeted indicator entity attribute changes in 2001–2021. The unit of indicators is millions. (c) 2019–2021 GDP prediction results (R-GCNs + XGBoost, RF, linear regression, LSTM, ST-GCN). (d) Accuracy assessment. Mean squared error (MSE) and mean absolute percentage error (MAPE).

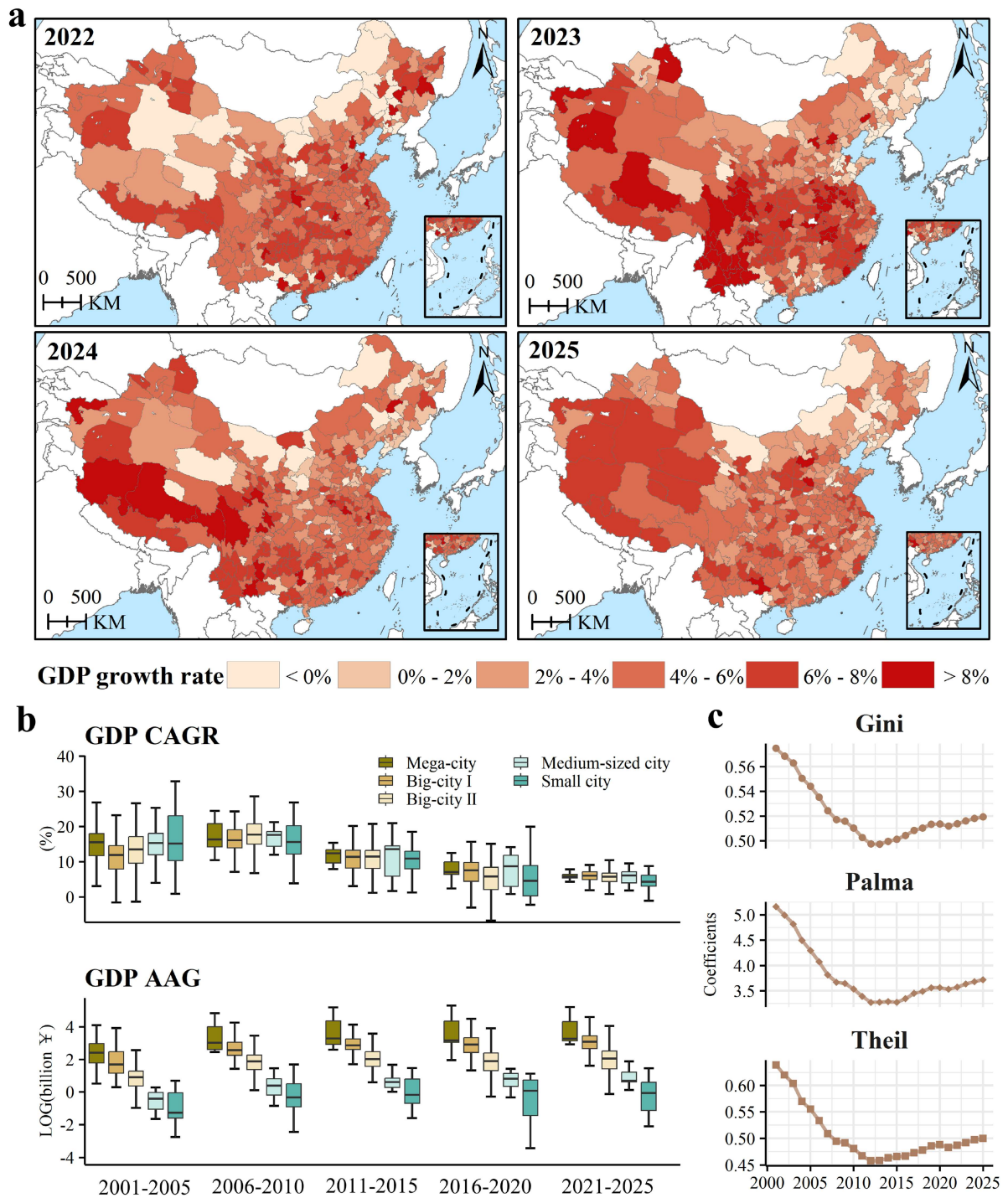


Figure 5. Prediction and analysis of GDP growth rates for each city from 2022 to 2025. (a) Prediction results of GDP growth rate in 2022–2025. (b) Comparison of GDP compound annual growth rate (CAGR) and average annual growth rate (AAG) in cities of different sizes. The boxes span from the first quartile (Q1, 25th percentile) to the third quartile (Q3, 75th percentile), and the error bar extend to 1.5 times the IQR. (c) Changes in the Economic Inequality indexes (Gini index, Palma index and Theil index).

Table 5. GDP growth rate from official data and predict data.

Year	National bureau of statistics	Predict	Absolute error	Relative error
2022	3.0%	3.26%	0.26%	8.67%
2023	5.2%	5.49%	0.29%	5.58%
2024	5.0%	4.99%	0.01%	0.20%
2025	/	4.39%	/	/

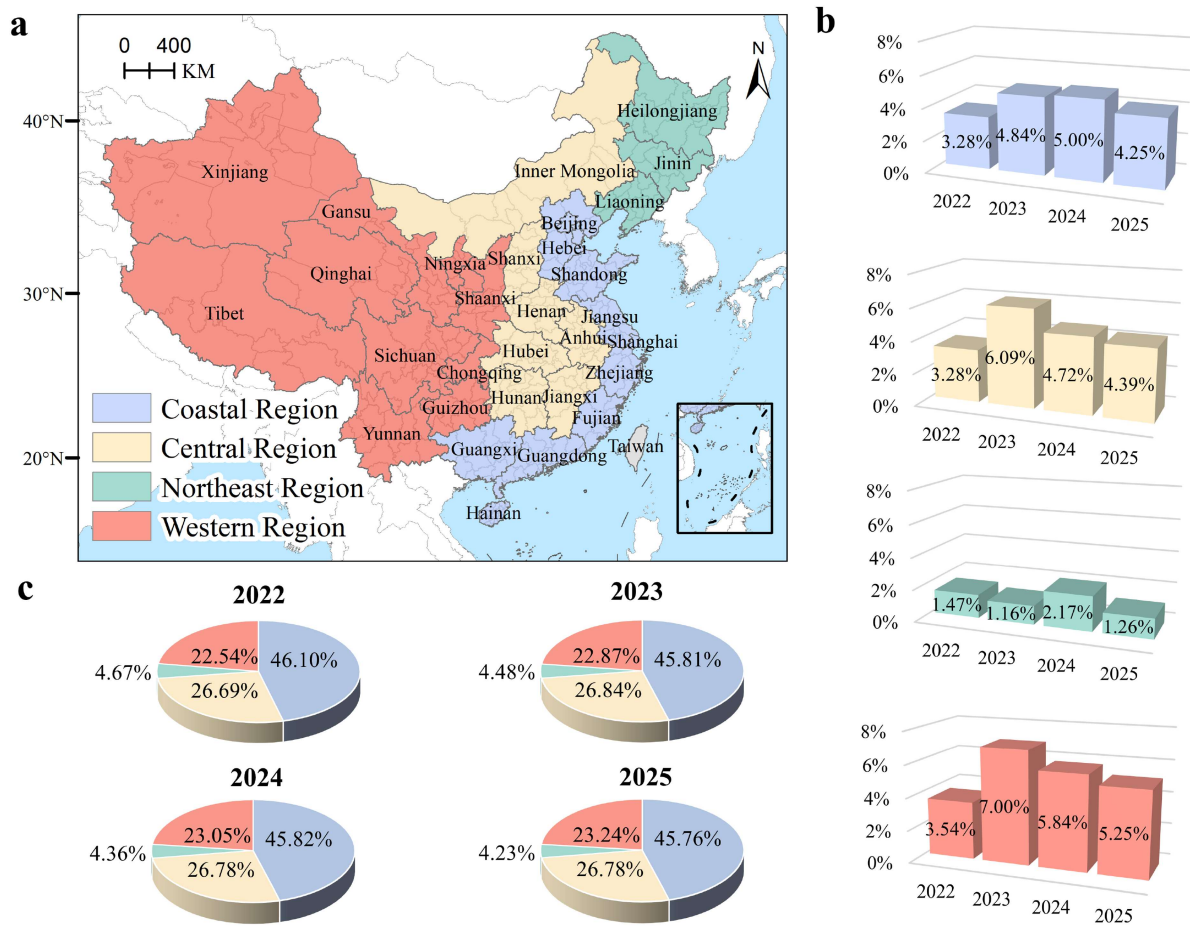


Figure 6. Economic development of China's four major regions from 2022 to 2025. (a) China's four major economic regions. (b) GDP growth rates of China's four major economic regions from 2022 to 2025. (c) The proportion of GDP contributed by the four major regions from 2022 to 2025.

remaining relatively stable (Figure 5b). Economic inequality indices (Gini, Palma, Theil) declined rapidly from 2000–2012, followed by a slow increase (2012–2025), and are projected to continue rising gradually unless targeted policies mitigate this inequality (Figure 5c). These findings provide critical insights into regional and city-level economic trends in China, guiding the formulation of policies to promote equitable and balanced development.

From the perspective of regional economic development, analysis of 2022–2024 data reveals that the western region exhibited the highest GDP growth rates, exceeding 5.5% in 2023 and 2024 (Figure 6b). The central region maintained relatively high growth, while the northeast region grew slowly at below 2.2%. Coastal areas sustained stable growth at 3%–5%, though their GDP share slightly declined (Figure 6c). Predictions for 2025 suggest that these trends will persist, with the western region continuing to lead, the northeast lagging, and coastal areas maintaining stable but modestly declining GDP contributions.

4 Discussion

4.1 Impact of external shocks on economic predictions

The R-GCNs + XGBoost model effectively predicts regional GDP growth using REST-KG embeddings, but struggles with sudden events like the 2020 COVID-19 pandemic. The model relies on historical patterns (2001–2021) limits its ability to adapt to unforeseen shocks, as seen in its underestimation of the 2020 GDP slowdown.

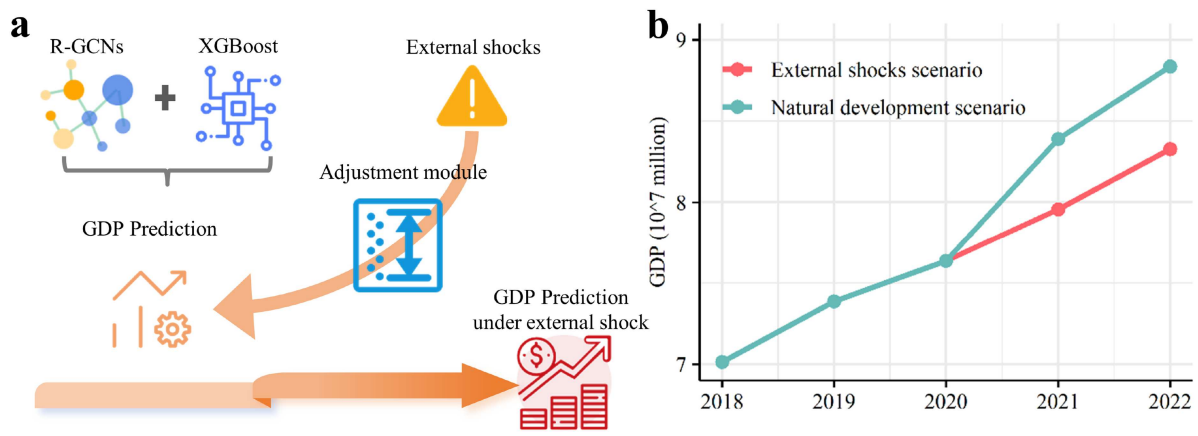


Figure 7. GDP prediction under external shock scenarios. (a) GDP prediction under external shock scenario. (b) China's GDP prediction for 2021 and 2022 under external shock scenarios.

To address this, we developed an external shock impact module, incorporating a binary feature (1 for shock years, e.g. 2020; 0 otherwise). Using 2001–2021 data with a sliding window (8-year input, 1-year output), we trained the module to adjust GDP growth predictions under shock scenarios. For example, training on 2012–2019 data with 2020's shock feature set to 1 refined 2020 predictions. For 2021–2022, we applied this module to adjust predictions when simulating continued pandemic impacts (feature = 1), leaving normal scenarios (feature = 0) unchanged. This approach improved prediction accuracy under shocks while preserving systemic economic relationships (Figure 7).

Future enhancements could include multi-level shock features (e.g. severity scores) to capture varying impact intensities, though data inconsistencies pose challenges. Integrating real-time data (e.g. mobility indices) and adaptive learning techniques could further enhance the module's responsiveness to sudden events, strengthening REST-KG's utility for policymaking in dynamic economic contexts.

4.2 Insights for regional economic policy

The regional economic spatiotemporal knowledge graph (REST-KG) provides novel insights into China's regional economic dynamics, revealing that primary GDP drivers—nighttime light, employment, consumption, and investment—exhibit temporal variations from 2001 to 2021, as validated by Structural Equation Modeling (SEM) with significant positive effects. This temporal decline in driver influence, attributed to technological innovation, globalization, and policy shifts, highlights the evolving nature of economic structures, where traditional indicators increasingly fail to capture emerging dynamics like digital economy integration. For instance, the high similarity of nighttime light to GDP underscores urbanization's role in growth, but its weakening correlation suggests a shift toward service-based economies, offering policy-makers evidence for transitioning from industrial to sustainable models.

Regionally, the findings show pronounced disparities: western and southeastern regions led with high GDP growth rate, the central region maintained robust growth, while the northeast grew slowly, with coastal areas contributing stably but declining in share. These patterns emphasize the success of policies like the Belt and Road Initiative in boosting western development, yet reveal challenges in revitalizing the northeast, informing targeted investments in infrastructure and innovation. Economic inequality, marked by inter-city disparities, narrowed from 2000–2012 but widened slowly thereafter, indicating that without interventions, gaps between medium and small cities could persist, undermining China's “Common Prosperity” agenda.

4.3 Limitations and future directions

REST-KG integrates heterogeneous entities and multidimensional relationships to model systemic economic interactions, accurately identifying GDP drivers and predicting future trends. However, it faces limitations that shape future planning. First, data comprehensiveness is limited by the need to unify disparate data

sources (e.g. the Statistical Yearbook and NPP-VIIRS from 2001 to 2021) across time and space. This ensures the reliability of the experiment but limits the inclusion of richer non-statistical data (e.g. web search indices or social media comments) that can capture dynamic socioeconomic trends. Second, the use of annual data limits high-frequency temporal granularity, missing short-term cyclical variations like monthly tourism or retail fluctuations.

Future planning involves integrating real-time data, such as mobility indices from smartphone tracking and e-commerce transactions from major platforms, to capture short-term economic fluctuations like monthly tourism or retail cycles, enabling more responsive policy adjustments. Additionally, incorporating digital indicators like internet penetration rates, green metrics such as carbon emissions or renewable energy adoption, and cultural indicators like tourist arrivals will allow finer analysis of economic transition features, such as shifts toward service-oriented or sustainable economies, and their underlying processes. Validating REST-KG on global datasets, such as EU NUTS regions or U.S. counties, will facilitate cross-country comparisons of regional economic differences, enhancing the model's scalability to diverse geographic and economic contexts. Furthermore, adopting advanced interpretability tools, such as SHAP values to quantify feature contributions or R-GCN attention weights to highlight influential relationships, will improve causal understanding beyond correlations, strengthen resilience against disruptions.

5 Conclusions

This study introduces the regional economic spatiotemporal knowledge graph (REST-KG), integrating traditional statistical indicators with non-statistical data to address limitations of conventional economic models reliant on static, single-source data and overcomes limitations of prior graph neural network predictions of geographic information, which often lack heterogeneous entities or multidimensional relationships. Grounded in the systemic nature of regional economies, REST-KG incorporates diverse entities and multidimensional relationships, enabling Relational Graph Convolutional Networks (R-GCNs) to capture systemic economic interactions through knowledge embedding, transforming heterogeneous data into actionable information. This information is further transformed into knowledge through indicator identification and economic forecasting knowledge reasoning tasks, thereby advancing dynamic regional economic analysis and providing strong support for informed policymaking and equitable development.

From 2001–2021 identified nighttime light, employment, consumption, and investment as primary drivers of GDP growth. The R-GCNs + XGBoost model, characterized by both these main drivers and knowledge embedding, accurately predicted GDP growth from 2022 to 2025, outperforming other models by leveraging systemic economic interactions and heterogeneous data for superior accuracy. To improve GDP growth forecasts under unexpected events, an external shock module was designed to enhance forecasting capabilities under extreme scenarios.

Over time, the influence of these primary drivers on GDP has gradually weakened due to changes in economic structure driven by technological progress, globalization, and policy changes. While most cities rely on these key drivers, economic patterns vary significantly across regions due to differences in structure, development paths, and resources. This highlights the need for region-specific policies. In China, economic inequality, characterized by inter-city economic development, narrowed significantly between 2000 and 2012 but has since slowly widened. Projections suggest that without targeted policies to promote equitable development, this inequality will further increase. REST-KG provides a powerful framework for analyzing economic trends, guiding targeted policymaking, and achieving balanced regional development.

Author contributions

Xiangdong Xu: Writing – review & editing, Writing – original draft, methodology, visualization, resources, conceptualization. **Limin Jiao:** Writing – review & editing, supervision, project administration, funding acquisition, conceptualization. **Gang Xu:** Writing – review & editing, conceptualization, visualization.

Disclosure statement

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Data availability statement

The data and codes that support the findings of this study are available on figshare.com with the identifier [DOI: <https://doi.org/10.6084/m9.figshare.29783996>].

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